

Decant Groceries and Label SOP

One episode decants three bulk packs of dry groceries into three jars. Each pack is opened, poured into a jar, and thrown out, then the jar is closed with its lid, labeled on its body, and stood in the crate. The three jars stand in the crate, one jar per slot.

The order never changes within a jar: stage, open, pour, discard the pack, close, label, shelve. No pack is lifted over a jar until its rolled top is unrolled and its pack mouth stands open, no lid goes on until the pour is finished, no label goes on until the jar is closed, and no jar goes in the crate until it is closed and labeled.

The left gripper stages the jars and steadies each jar while the right gripper works on it. The right gripper takes the packs, pours, discards the empty packs, closes the lids, applies the labels, and shelves the finished jars. Neither gripper crosses to the other side of the table.

A jar is never lifted while it is open and holding product, nothing is turned in the air, no gripper opens anywhere but over the spot the thing in it is going to, and product that leaves a pack goes into a jar and nowhere else.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the pack row at the back right, the crate at the back center, the jar row at the front left, the lid rest right of the working area, the label board in front of the working area, and the waste bin at the front right corner.
3. The working area at the center of the table is visible from above, so every pour can be seen going into the jar mouth and any product that lands on the table can be seen.
4. Both arms are at home with grippers open.
5. The table is bare apart from the packs, the jars, the lids, the label board, the crate, and the waste bin.
6. The left arm reaches the jar row and the working area. The right arm reaches the pack row, the lid rest, the label board, the working area, the crate, and the waste bin.

Materials checklist

1. Three **bulk packs** of dry groceries stand in a row at the back right, each closed by a rolled top. Each pack holds one jar fill or less, so a whole pack empties into one jar without filling it to the jar mouth.
2. The three packs stand in the same left to right order as the three labels on the label sheet, so the next pack always matches the next label.
3. Three empty **jars** stand upright in a row at the front left, mouths open and bases flat.
4. Three **lids**, one per jar, lie in a row on the lid rest, interior down against the clean rest, so a lid is lifted straight onto a jar mouth and never has to be turned over.
5. One **label sheet** carrying three written labels lies on the label board, written side up, held flat by the board so no gripper has to hold it. Each label names the product in one pack.
6. One **crate** sits at the back center with its slots empty, each slot deep enough that a jar stands in it without leaning.
7. One open **waste bin**, empty, at the front right corner, its rim clear of the label board.
8. The working area at the center of the table is clear and dry.

Workspace layout

- **Pack row:** back right, the three closed bulk packs standing in a row.
- **Crate:** back center, its slots empty.
- **Jar row:** front left, the three empty jars standing upright.
- **Working area:** the center of the table, where every open, pour, close, and label happens.
- **Lid rest:** right of the working area, the three lids lying interior down.
- **Label board:** in front of the working area, the label sheet held flat and written side up.
- **Waste bin:** front right corner, open and empty.

Everything the right arm works on is staged on the right or in the center. The jars, the only things the left arm carries, are staged on the left. No carried thing crosses the midline.

Vocabulary

- **Arm assignment:** the left gripper stages the jars and steadies a jar while the right gripper works on it. The right gripper takes the packs, pours, discards the empty packs, closes the lids, applies the labels, and shelves the jars. Neither gripper takes the other's zones.
- **Bulk pack:** one flexible pack of dry groceries, closed by a rolled top, holding one jar fill or less. Its **pack base** is the closed bottom end, opposite the rolled top.
- **Rolled top:** the top of a bulk pack folded over on itself to hold it closed.
- **Pack mouth:** the opening of a bulk pack once the rolled top is unrolled and its two sides are pulled apart.

- **Product:** the dry groceries inside a bulk pack. Product goes from a pack into a jar and nowhere else.
- **Jar:** one rigid storage jar that receives the product from one pack. Its **jar mouth** is the threaded opening at the top and its **jar body** is the straight side wall between the base and the jar mouth.
- **Lid:** the screw lid that closes one jar. It is held only by its **lid rim**, the outer edge. A gripper never closes on the interior of a lid, and no lid is ever turned over in the air.
- **Closed jar:** the lid turned onto the jar mouth until it stops turning, sitting level with no gap at the jar mouth.
- **Turn and regrip:** how a lid is closed. Turn the lid clockwise as far as the wrist goes, open the gripper, bring the wrist back, close on the lid rim again, and turn again. Repeat until the lid stops turning. A lid is never closed in one continuous twist.
- **Label:** one written adhesive label naming one product. Its **label face** is the written side.
- **Labeled jar:** a closed jar carrying one label pressed flat on the jar body, label face out and the writing upright.
- **Crate slot:** one of the upright positions in the crate, each one holding one jar.
- **Open slot:** a crate slot that does not yet hold a jar. A slot that holds a jar is never used again in the episode.
- **Steady hold:** the left gripper closed on the jar body with the jar upright and its base flat on the table, held for the whole of a pour, a close, or a label press.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Lower in:** carrying level over the crate slot or the waste bin, lowering straight down until the thing is resting, and releasing once it is resting. Nothing is dropped in from above the rim.
- **Settled:** the thing stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, or tipping.
- **Release point:** over the working area, over the crate slot, or over the waste bin the thing in the gripper is going to. A gripper opens nowhere else.
- **Spill:** any product outside a bulk pack, a jar, or the waste bin.

Steps

Steps 1 to 7 fill one jar from one pack. Run them three times, taking the packs left to right along the pack row and the jars left to right along the jar row, and standing each finished jar in an open crate slot. Step 8 ends the episode once the third jar is in the crate.

Step 1: Stage a jar and a pack in the working area

Goal: one empty jar standing upright in the working area with one closed pack lying flat beside it on its right.

- The **left gripper** closes on the **jar body** of the leftmost jar still in the jar row, lifts it straight up, carries it upright to the **working area**, lowers it until its base is flat, and releases.
- The **right gripper** closes on the leftmost **bulk pack** still in the pack row around its middle, lifts it straight up, carries it level to the working area, lays it flat beside the jar on the jar's right with the rolled top pointing toward the jar, and releases.

Take one jar and one pack. Never lift two jars or two packs in one grasp, and never drag either one across the table.

Check: one jar stands upright in the working area with its base flat and its mouth open, and one closed pack lies flat beside it, clear of it, with the rolled top pointing toward the jar.

Step 2: Open the bulk pack

Goal: the rolled top unrolled and the pack mouth standing open, with the pack still flat on the table.

- The **left gripper** closes on the **pack base** and holds the pack flat on the table.
- The **right gripper** closes on the outer edge of the **rolled top** and unrolls it away from the pack base until the top lies flat and the pack mouth is clear.
- If the rolled top does not come free in one unroll, take the outer edge again and unroll a second time.
- The **right gripper** then closes on one side of the **pack mouth** and pulls it away from the other side until the mouth stands open.
- Both grippers release.

The pack stays flat on the table for the whole of this step. Never lift a pack to open it, and never lift a closed pack over a jar.

Check: the rolled top lies flat, the pack mouth stands open, the pack is still flat on the table, and no product has come out of it.

Step 3: Decant the pack into the jar

Goal: all of the product in the jar, sitting below the jar mouth, and the pack empty.

- The **left gripper** takes a **steady hold** on the jar body.
- The **right gripper** closes on the **pack base**, lifts the pack clear of the table, and brings the pack mouth over the jar mouth, centered on it and just above it.
- Tilt the pack toward the jar mouth in one slow motion and hold the pack mouth centered over the jar mouth while the product runs in.
- If the product stops running while product is still in the pack, bring the pack upright, then tilt again.

- Once no product is left in the pack, bring the pack upright and hold it clear of the jar. The **left gripper** keeps its steady hold until the pack is clear.

The jar stays on the table through the whole pour, so never lift the jar to meet the pack and never pour from a height above the jar mouth. If the product rises to the jar mouth while product is still in the pack, stop pouring and end the episode as an overfilled pack (see Failures that are not violations).

Check: the pack is empty, the product sits below the jar mouth with the thread clear, and there is no product on the table, in the crate, or anywhere outside the jar.

Step 4: Discard the empty pack

Goal: the empty pack in the waste bin and the working area holding only the open jar.

- The **right gripper**, still holding the empty pack, carries it level to the **waste bin**, lowers it in, and releases.
- The **left gripper** releases the jar once the pack is clear of the working area.

Check: the waste bin holds the empty pack, the working area holds only the open jar standing upright, and no product came out of the pack on the way.

Step 5: Close the jar

Goal: a closed jar, its lid seated level and turned until it stops.

- The **left gripper** takes a **steady hold** on the jar body.
- The **right gripper** closes on the **lid rim** of the leftmost lid still on the lid rest, lifts it straight up, carries it level over the jar, and lowers it onto the **jar mouth**, seated square and sitting level.
- Close it with **turn and regrip**, repeating until the lid stops turning.
- Release the right gripper, then the left.

Hold the lid by its rim only. The interior of a lid touches the product, so no gripper closes on it and no lid is turned over in the air. Never lift the jar while it is open.

Check: the lid sits level on the jar mouth with no gap, it does not turn further, and it does not lift off when the jar body is nudged.

Step 6: Label the jar

Goal: one label pressed flat on the jar body, naming the product in that jar.

- The **left gripper** takes a **steady hold** on the jar body, jar upright.
- The **right gripper** closes on the corner of the **label** whose label face names the product just decanted, and peels it off the label sheet in one motion with the label face out.

- Bring the label to the side of the **jar body** facing the operator, label face out and the writing upright.
- Press it onto the jar body centered on the jar body height, then press along it from one edge to the other until it lies flat with no lifted corner.
- Release both grippers.

Read the label face before peeling it. If a label tears or folds onto itself, set it aside on the label board and peel the next label naming the same product.

Check: the jar is closed and carries one label, the label face names the product in that jar, the writing is upright, the label lies flat with no lifted corner, and the label face points toward the operator.

Step 7: Shelf the jar in the crate

Goal: the closed and labeled jar standing upright in an open crate slot with its label face toward the front of the table.

- The **right gripper** closes on the **jar body** without turning the jar, lifts it straight up, carries it upright to the **crate**, and lowers it into an **open crate slot** until its base sits flat.
- Release once the jar is resting.

The jar is carried in the orientation it was labeled in, so the label face still points toward the front of the table when it is set down. Nothing is turned in the air. Never press a jar down into a slot and never lower a jar into a slot that already holds one.

Check: the jar stands upright in a slot that held no jar before, with its base flat, it does not lean, and its label face points toward the front of the table.

Step 8: Confirm the three jars and end the episode

- Confirm the crate holds three closed and labeled jars, one per slot, each standing upright with its label face toward the front of the table, and confirm the jar row, the pack row, and the lid rest are empty, the waste bin holds three empty packs, and the working area is bare.
- Leave the scene untouched for 3 seconds with the crate, the working area, and the waste bin all in frame. Nothing is held or moved during the hold.
- Return both arms home, clear of the crate and the working area, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the three jars out of the crate and set the empty crate back at the back center with its slots clear.

2. Open each jar counterclockwise, set its lid aside, and empty the product back into a fresh bulk pack. Roll the top of each pack closed.
3. Peel the label off each jar body and set the jars aside. Replace a jar whose thread will not take a lid.
4. Stand the three fresh packs in a row at the back right in the same left to right order as the labels on the new label sheet.
5. Stand the three empty jars upright in a row at the front left, mouths open and bases flat.
6. Wipe each lid clean and lay the three lids interior down in a row on the lid rest.
7. Put a fresh label sheet on the label board, written side up and held flat, with one label per pack.
8. Empty the waste bin and set it back at the front right corner.
9. Wipe any product off the table and confirm the surface is clean and dry. A crumbed table lets a jar slide under a steady hold.
10. Vary which product sits in which position along the pack row between episodes rather than running the same order every time. The count stays at three packs, three jars, three lids, and three labels.

SOP violations

These are operator actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every operator violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because an operator broke a rule.

Violations

Violation: Wrong order

- **Visible cue:** a pack is lifted over a jar before its pack mouth is open, a lid goes on before the pour is finished, a label goes on before the jar is closed, a jar goes in the crate before it is labeled, or the empty pack is still in the working area when the lid goes on.
- **SOP rule broken:** Steps 1 to 7, the order within a jar never changes.
- **Coaching note:** stage, open, pour, discard, close, label, shelf. Finish the step you are in before you start the next one.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes a pack, a lid, a label, or carries a jar to the crate or the waste bin; or the right gripper takes a jar out of the jar row.
- **SOP rule broken:** Steps 1 to 7, the left gripper stages and steadies the jars and the right gripper does the packs, the lids, the labels, and the shelving.
- **Coaching note:** each arm stays on its own side. Nothing crosses the midline.

Violation: More than one jar or pack handled at once

- **Visible cue:** two or more jars are lifted from the jar row in one grasp, or two or more packs are lifted from the pack row in one grasp.
- **SOP rule broken:** Step 1, one jar and one pack per cycle.
- **Coaching note:** one jar, one pack, one label, one lid, one slot. Never bundle a cycle.

Violation: Pack lifted to open it

- **Visible cue:** the pack leaves the table during the unroll, the rolled top is worked open in the air, or the pack mouth is pulled open while the pack hangs from a gripper.
- **SOP rule broken:** Step 2, the pack stays flat on the table for the whole of the open.
- **Coaching note:** hold the pack base down and unroll against the table. Nothing opens in the air.

Violation: Poured while the pack was still closed

- **Visible cue:** the pack is lifted over the jar with the rolled top still rolled, or with the pack mouth never pulled open.
- **SOP rule broken:** Steps 2 and 3, the pack mouth stands open before the pack goes over the jar.
- **Coaching note:** open it flat on the table, then lift it. Never carry a closed pack over a jar.

Violation: Jar not steadied during a pour, close, or press

- **Visible cue:** the left gripper is off the jar body while the product runs, while the lid is turned, or while the label is pressed; or the jar tips, slides, or lifts off the table during any of them.
- **SOP rule broken:** Steps 3, 5, and 6, the left gripper holds the jar upright with its base flat for the whole of each.
- **Coaching note:** the steady hold goes on before the right gripper starts and comes off after it finishes.

Violation: Poured from too high or off center

- **Visible cue:** the pack mouth is well above the jar mouth or is not over it when the tilt starts, or it drifts off the jar mouth while the product runs.
- **SOP rule broken:** Step 3, bring the pack mouth just above the jar mouth and centered on it, and hold it there for the whole pour.

- **Coaching note:** get low and centered before you tilt, and stay there until it stops running.

Violation: Spill

- **Visible cue:** product lands on the table, in the crate, on the label board, or anywhere outside the jar and the waste bin at any point in the episode.
- **SOP rule broken:** Step 3, all of the product goes into the jar.
- **Coaching note:** center the pack mouth and tilt in one slow motion so the product runs rather than drops.

Violation: Kept pouring into a full jar

- **Visible cue:** the product stands level with or above the jar mouth at the end of the pour, or the operator goes on tilting after the product has risen to the jar mouth.
- **SOP rule broken:** Step 3, the product sits below the jar mouth with the thread clear.
- **Coaching note:** watch the product rise during the pour and bring the pack upright as soon as it reaches the jar mouth.

Violation: Product left in the pack

- **Visible cue:** the pack is set down or thrown out with product still in it while the product in the jar is still below the jar mouth.
- **SOP rule broken:** Step 3, the pack is empty at the end of the pour.
- **Coaching note:** bring it upright and tilt again until nothing more runs out.

Violation: Empty pack not discarded

- **Visible cue:** the lid goes on, the label goes on, or the jar is shelved while the empty pack is still in the working area, on the pack row, or in the crate.
- **SOP rule broken:** Step 4, the empty pack goes in the waste bin before the jar is closed.
- **Coaching note:** the pack goes straight from the pour to the bin. Do not put it down first.

Violation: Lid mishandled

- **Visible cue:** a gripper closes on the interior of a lid instead of the lid rim, or a lid is turned over between the lid rest and the jar mouth.
- **SOP rule broken:** Step 5, a lid is held by its rim only and is never turned over in the air.
- **Coaching note:** the interior touches the product. Take the rim and carry it the way it lies.

Violation: Jar not closed

- **Visible cue:** the lid rests on the jar mouth without being turned, sits at an angle, shows a gap, or lifts off when the jar is moved.
- **SOP rule broken:** Step 5, seat the lid level and turn and regrip until it stops turning.
- **Coaching note:** seat it square first, then turn and regrip. One twist is never enough.

Violation: Lid twisted without regripping

- **Visible cue:** the wrist runs to its limit and stalls while still closed on the lid, or the gripper drags on the lid rim instead of opening and taking a fresh hold.
- **SOP rule broken:** Step 5, open the gripper, bring the wrist back, and close on the lid rim again between turns.
- **Coaching note:** turn, let go, come back, take it again. The wrist does not have the range to close a lid in one go.

Violation: Wrong label applied

- **Visible cue:** the label face on the jar names a product other than the one decanted into that jar.
- **SOP rule broken:** Step 6, peel the label whose label face names the product just decanted.
- **Coaching note:** read the label face before you peel it and confirm it matches what went in.

Violation: Label crooked, off center, or lifting

- **Visible cue:** the writing on the label face is not upright, the label sits high or low on the jar body rather than centered on its height, or a corner of it stands off the jar.
- **SOP rule broken:** Step 6, the label goes on with the writing upright, centered on the jar body height, and pressed flat with no lifted corner.
- **Coaching note:** line the writing up before it touches, then press from one edge to the other.

Violation: Label applied to an open jar

- **Visible cue:** a label is peeled or pressed while the jar is not yet a closed jar.
- **SOP rule broken:** Steps 5 and 6, the jar is closed before it is labeled.
- **Coaching note:** close it, then label it. Never the other way round.

Violation: Jar shelved in a slot that already holds one

- **Visible cue:** the jar is lowered into a crate slot that already holds a jar, or it is stood on top of, wedged beside, or leaned against a jar already in the crate.
- **SOP rule broken:** Step 7, the jar goes into an open crate slot, one jar per slot.
- **Coaching note:** find a slot that is still empty before you lower the jar.

Violation: Jar shelved leaning, pressed down, or off zone

- **Visible cue:** the jar is released outside the crate, left in the working area, set down leaning, or pressed down into a slot after it is already resting.
- **SOP rule broken:** Step 7, lower the jar into the slot until its base sits flat and release once it is resting.
- **Coaching note:** take it all the way down, then let go. Leave it how it lands.

Violation: Label face turned away from the front

- **Visible cue:** the jar is turned in the gripper between the label press and the crate, or the label face on the shelved jar points to the back or the side of the table.
- **SOP rule broken:** Step 7, the jar is carried in the orientation it was labeled in and nothing is turned in the air.
- **Coaching note:** the orientation is set when the label goes on. Carry it straight back and do not twist.

Violation: Jar lifted while open

- **Visible cue:** the jar leaves the table between the pour and the close, or an open jar holding product is carried, tipped, or slid.
- **SOP rule broken:** Steps 3 to 5, a jar is never lifted while it is open and holding product.
- **Coaching note:** the jar stays on the table until its lid is on.

Violation: Dragged instead of lifted

- **Visible cue:** a jar, a pack, or a lid slides along the table still touching it, with no daylight under it in transit.
- **SOP rule broken:** Steps 1 to 7, lift straight up until there is daylight before moving sideways.
- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Regripped in the air

- **Visible cue:** the gripper shifts, rolls, or seats its hold again on a jar, a pack, or a label without setting it down first. The turn and regrip cycle in Step 5 is not this.
- **SOP rule broken:** Steps 1 to 7, set it down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Dropped instead of lowered

- **Visible cue:** a gripper opens above the crate or the waste bin and the thing falls the rest of the way in, or a jar is released before its base is resting.
- **SOP rule broken:** Steps 1, 4, and 7, release only once the thing is resting.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the table, over the crate when it is carrying a pack, over the waste bin when it is carrying a jar, or over open space.
- **SOP rule broken:** Steps 1 to 7, a gripper opens over the spot the thing is going to and nowhere else.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides a jar, a pack, a lid, or a label into position after releasing it, instead of lifting it clear and placing it again.
- **SOP rule broken:** Steps 1 to 7, correct a bad placement by lifting the thing clear and placing it again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Repeated fiddling

- **Visible cue:** more than about two small corrections to settle a jar, a pack, a lid, or a label rather than settling it in one or two.
- **SOP rule broken:** Steps 1 to 7, each placement settles in one or two small corrections.
- **Coaching note:** the approach is off, so the first placement lands wide. Tighten the approach rather than correcting after.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** a jar, a pack, a lid, or a label falls to the table or the floor short of its spot, a jar in the crate is knocked over, the label board or the waste bin is struck and shifts, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 7, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a jar still out of the crate, a jar unlabeled or open, a pack still on the table, a lid still on the lid rest, product on the table, an arm away from home, an arm still over the crate or the working area, or recording stopped before 3 seconds of a still scene.
- **SOP rule broken:** Step 8, confirm the three jars and the empty zones; leave the scene untouched for 3 seconds; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm and hold first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not the operator's fault. Log them as system issues, discard the episode, and never use them to coach the operator.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.

- **Overfilled pack:** the product rises to the jar mouth while product is still in the pack, so one pack will not fit in one jar. Repack to one jar fill or less before the next episode.
- **Defective bulk pack:** a torn or already open pack, or one whose rolled top will not unroll. Replace it before the next episode.
- **Defective jar or lid:** a cracked jar or a thread that will not take a lid. Replace it before the next episode.
- **Defective label:** a label with no adhesive left, writing that cannot be read, or a label that tears on every peel. Replace the sheet before the next episode.
- **Pack and label out of order:** the pack row order does not match the label sheet order when the episode begins, so no label matches the product decanted. Restage before the next episode.
- **Jar slides** under a correct steady hold because the table is slick or crumbed. Wipe and dry the table before the next episode.

Annotation subtasks (from SOP)

1. Stage one jar and one pack in the working area
2. Open one bulk pack
3. Decant one pack into one jar
4. Discard one empty pack
5. Close one jar with its lid
6. Apply one written label to one jar
7. Shelf one closed and labeled jar in the crate
8. Confirm the three jars and end the episode